# Saswad farmland (season control)

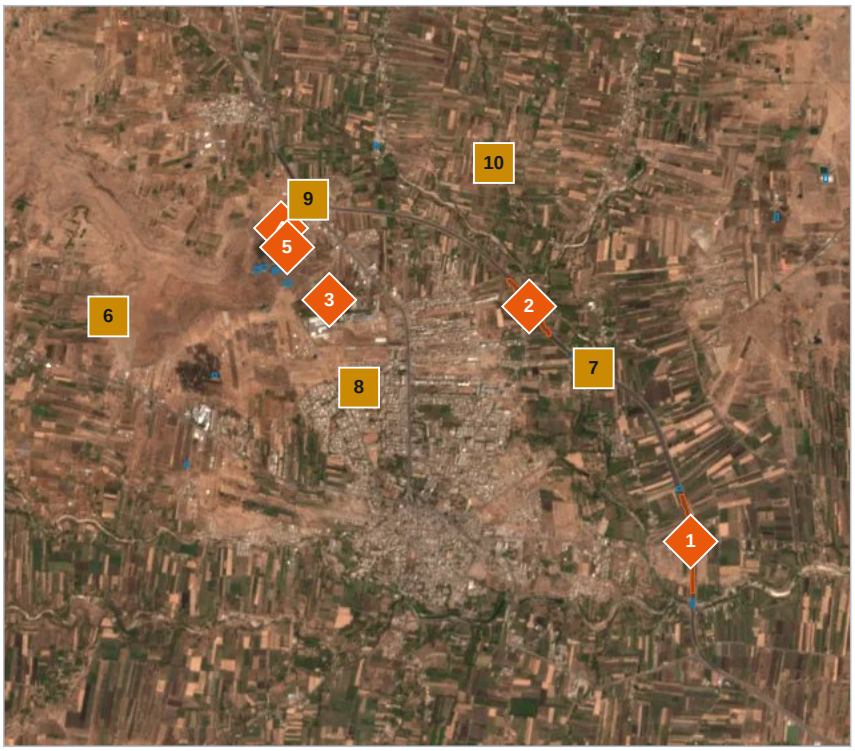
Farmland near Saswad, Pune district, Maharashtra | Before - Sentinel-2 - 2024-11-19 to After - Sentinel-2 - 2025-03-14

## What we found

We found 22 real changes covering 11.89 hectares. The biggest and strongest (rank 1) is new construction: 2.42 ha at 18.3437, 74.0488.

By priority: 0 Critical, 5 High, 5 Medium, 12 Low.

The detector first flagged 3,083 candidate areas. The rest were removed as false alarms: change types this analysis does not track, seasonal crop and vegetation cycles, tiny specks and noise.




Numbers = rank of the 10 highest-priority changes. Outline colour = priority.

### Changes by priority

| | |
|---|---|
| **Critical** | 0 |
| **High** | 5 |
| **Medium** | 5 |
| **Low** | 12 |

### Totals

Area changed: 11.89 ha

Cropland affected: 3.83 ha

Built-up land affected: 0.38 ha

## How to read this report

| | |
|---|---|
| **Rank** | Every change has a number. Rank 1 is the highest priority: the largest and strongest change. Priority = size x strength of the change. |
| **Priority colour and shape** | Critical (octagon): score 20,000 or more (about 2 ha of strong change); High (diamond): score 5,000 or more; Medium (square): score 1,500 or more; Low (circle): score below 1,500 (small or weak). |
| **Confidence** | How sure the system is that the change is real, from 0 to 100 %. Changes marked 'needs a human check' sit between 50 % and 70 %. |
| **What changed** | New construction, vegetation cleared, vegetation growth, new water / flooding, water receded, or unexplained change. These come from rules on the satellite colours, not from knowing what is actually on the ground. |
| **Raw detections** | Everything the detector first flagged. Most are not real (clouds, crop seasons, shadows, specks), so only changes that pass the filters are listed here. |

## The 10 highest-priority changes

Start here. Each card says where the change is, how big it is, why it was flagged and what to do.

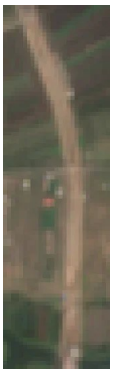
1

### New construction

Rank 1 of 22 | High priority

| | |
|---|---|
| **Size** | 2.42 ha (24,200 m2), about 3 football pitches |
| **Confidence** | 95 % |
| **Location** | 18.34368, 74.04882 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.24), vegetation index fell (-0.06), water index rose (+0.08), SWIR/blue ratio (paved surfaces are flatter) rose (+0.55) |
| **What to do** | Check soon. A substantial change that is probably real. |

Before



After



2

### New construction

Rank 2 of 22 | High priority

| | |
|---|---|
| **Size** | 1.99 ha (19,850 m2), about 3 football pitches |
| **Confidence** | 96 % |
| **Location** | 18.35975, 74.03727 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.24), vegetation index fell (-0.07), water index rose (+0.07), SWIR/blue ratio (paved surfaces are flatter) rose (+0.61) |
| **What to do** | Check soon. A substantial change that is probably real. |

Before



After



3

### New construction

Rank 3 of 22 | High priority

| | |
|---|---|
| **Size** | 1.67 ha (16,650 m2), about 2 football pitches |
| **Confidence** | 94 % |
| **Location** | 18.36020, 74.02298 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.21), vegetation index fell (-0.14), water index rose (+0.13), SWIR/blue ratio (paved surfaces are flatter) rose (+0.72) |
| **What to do** | Check soon. A substantial change that is probably real. |

Before



After



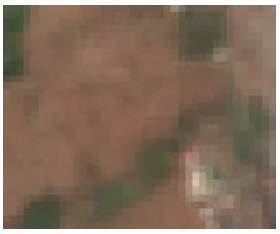
4

### New construction

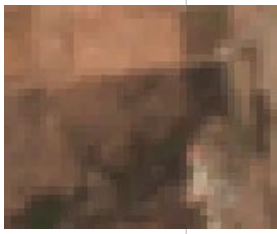
Rank 4 of 22 | High priority

| | |
|---|---|
| **Size** | 1.64 ha (16,400 m2), about 2 football pitches |
| **Confidence** | 90 % |
| **Location** | 18.36511, 74.01953 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.18), vegetation index fell (-0.15), water index rose (+0.14), SWIR/blue ratio (paved surfaces are flatter) rose (+0.18) |
| **What to do** | Check soon. A substantial change that is probably real. |

Before



After



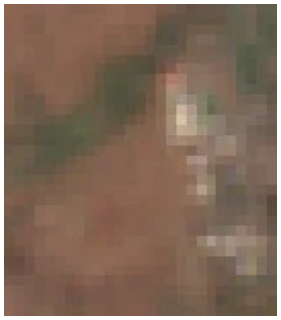
5

## New construction

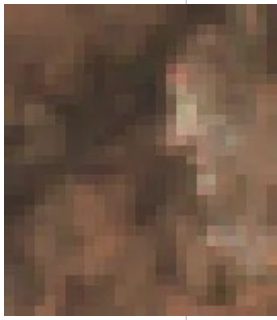
Rank 5 of 22 | High priority

| | |
|---|---|
| **Size** | 0.85 ha (8,550 m2) |
| **Confidence** | 93 % |
| **Location** | 18.36380, 74.01996 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.19), vegetation index fell (-0.14), water index rose (+0.14), SWIR/blue ratio (paved surfaces are flatter) rose (+0.19) |
| **What to do** | Check soon. A substantial change that is probably real. |

Before



After



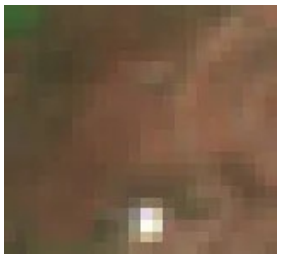
6

## New construction

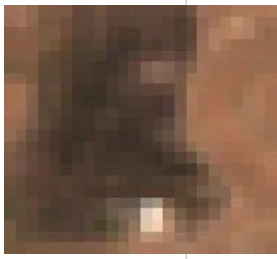
Rank 6 of 22 | Medium priority

| | |
|---|---|
| **Size** | 0.42 ha (4,200 m2) |
| **Confidence** | 95 % |
| **Location** | 18.35907, 74.00718 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.24), vegetation index fell (-0.16), water index rose (+0.19), SWIR/blue ratio (paved surfaces are flatter) fell (-0.45) |
| **What to do** | Check when you can, for example on a routine inspection. |

Before



After



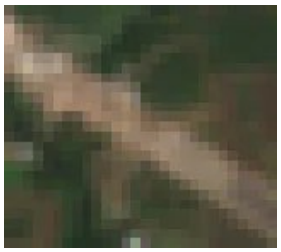
7

## New construction

Rank 7 of 22 | Medium priority

| | |
|---|---|
| **Size** | 0.44 ha (4,400 m2) |
| **Confidence** | 94 % |
| **Location** | 18.35552, 74.04189 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.25), SWIR/blue ratio (paved surfaces are flatter) rose (+0.77) |
| **What to do** | Check when you can, for example on a routine inspection. |

Before



After



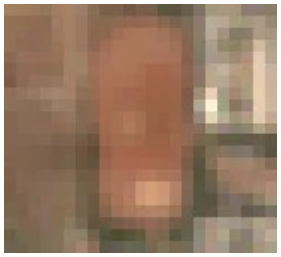
8

## New construction

Rank 8 of 22 | Medium priority

| | |
|---|---|
| **Size** | 0.24 ha (2,400 m2) |
| **Confidence** | 99 % |
| **Location** | 18.35420, 74.02511 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index fell (-0.24), vegetation index fell (-0.11), water index rose (+0.28), SWIR/blue ratio (paved surfaces are flatter) fell (-3.23) |
| **What to do** | Check when you can, for example on a routine inspection. |

Before



After



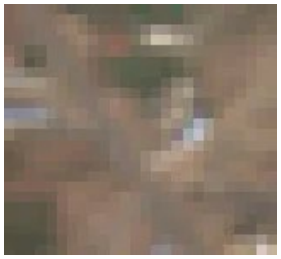
9

## New construction

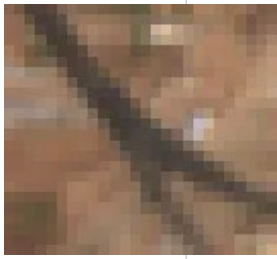
Rank 9 of 22 | Medium priority

| | |
|---|---|
| **Size** | 0.35 ha (3,500 m2) |
| **Confidence** | 94 % |
| **Location** | 18.36705, 74.02147 (open in Google Maps) |
| **What it is** | Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete. |
| **Why flagged** | built-up index rose (+0.24), vegetation index fell (-0.09), water index rose (+0.10), SWIR/blue ratio (paved surfaces are flatter) rose (+0.47) |
| **What to do** | Check when you can, for example on a routine inspection. |

Before



After



10

## Water receded

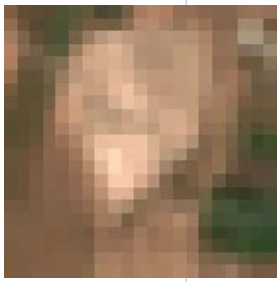
Rank 10 of 22 | Medium priority

| | |
|---|---|
| **Size** | 0.16 ha (1,600 m2) |
| **Confidence** | 99 % |
| **Location** | 18.36956, 74.03475 (open in Google Maps) |
| **What it is** | Water that was there before has gone. |
| **Why flagged** | vegetation index rose (+0.06), water index fell (-0.26), SWIR/blue ratio (paved surfaces are flatter) rose (+0.70) |
| **What to do** | Check when you can, for example on a routine inspection. |

Before



After



## All 22 changes found

Every change that survived the false-alarm filters, in priority order (rank 1 is the highest). A * after the confidence means it needs a human check. The CSV and GeoPackage downloads hold the same list with every attribute.

| Rank | Priority | What changed | Size (ha) | Confidence | Latitude | Longitude |
|---|---|---|---|---|---|---|
| 1 | High | New construction | 2.42 | 95 % | 18.34368 | 74.04882 |
| 2 | High | New construction | 1.99 | 96 % | 18.35975 | 74.03727 |
| 3 | High | New construction | 1.67 | 94 % | 18.36020 | 74.02298 |
| 4 | High | New construction | 1.64 | 90 % | 18.36511 | 74.01953 |
| 5 | High | New construction | 0.85 | 93 % | 18.36380 | 74.01996 |
| 6 | Medium | New construction | 0.42 | 95 % | 18.35907 | 74.00718 |
| 7 | Medium | New construction | 0.44 | 94 % | 18.35552 | 74.04189 |
| 8 | Medium | New construction | 0.24 | 99 % | 18.35420 | 74.02511 |
| 9 | Medium | New construction | 0.35 | 94 % | 18.36705 | 74.02147 |
| 10 | Medium | Water receded | 0.16 | 99 % | 18.36956 | 74.03475 |
| 11 | Low | New construction | 0.28 | 89 % | 18.36237 | 74.01803 |
| 12 | Low | Water receded | 0.15 | 98 % | 18.36852 | 74.05849 |
| 13 | Low | New construction | 0.24 | 86 % | 18.36225 | 74.01926 |
| 14 | Low | Water receded | 0.12 | 98 % | 18.37075 | 74.02635 |
| 15 | Low | New construction | 0.21 | 84 % | 18.36130 | 74.01998 |
| 16 | Low | New construction | 0.12 | 95 % | 18.33941 | 74.04890 |
| 17 | Low | New construction | 0.09 | 97 % | 18.35641 | 74.04048 |
| 18 | Low | New construction | 0.15 | 84 % | 18.36588 | 74.05491 |
| 19 | Low | New construction | 0.09 | 94 % | 18.34732 | 74.04801 |
| 20 | Low | New construction | 0.09 | 91 % | 18.35501 | 74.01484 |
| 21 | Low | New construction | 0.09 | 88 % | 18.36344 | 74.01915 |
| 22 | Low | New construction | 0.09 | 85 % | 18.34894 | 74.01279 |

## Data sources

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0

## Limits - what this report cannot tell you

- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero
- This report points to where to look. Confirm important changes on the ground before acting.